



locally. OEMs are now more cautious and thorough in testing components, even locally sourced, to ensure they meet stringent quality standards.

How can MSMEs increase their contribution to what major investment-heavy OEMs are trying to achieve?

MSMEs should invest in R&D and align with global standards to improve their offerings. The trend is shifting from having everything in-house to leveraging external expertise. By adopting proven technologies and innovating, MSMEs can enhance their value proposition. There is a governmental push for higher local content, which will increase from current levels to mandate higher domestic production percentages in the coming years. This shift creates a substantial opportunity for MSMEs to scale up their capabilities and meet the increasing demand for locally produced components. Additionally, testing facilities in India are becoming more sophisticated and accessible, further supporting the development of high-quality, locally manufactured battery components.

Given the significant downturn in the electric two-wheeler and three-wheeler market from March to April, with sales dropping over 50% and the segment decreasing by more than 70%, should MSMEs view this as a critical area of focus?

Yes, this should be a focus area. The downturn is temporary, reflecting short-term challenges rather than long-term issues. The shift towards electric vehicles is inevitable and expected to encompass 70 to 80% of the two-wheeler and three-wheeler market in the next three to four years. Regulatory changes and evolving consumer preferences are driving this shift. As electric vehicle (EV) costs become competitive with

traditional vehicles and the long-term viability of EVs increases, the market is poised for recovery. This presents an opportunity for MSMEs to position themselves strategically for upcoming demand, particularly in high-voltage systems and components.

With academia struggling to engage effectively with large and small companies, what role do you see for academic institutions in developing high-voltage batteries, and how can this collaboration be extended to MSMEs?

Academic institutions play a crucial role in addressing the skill gaps in the EV sector. There is a substantial need for programmes that go beyond theoretical knowledge to provide hands-on experience with high-voltage systems and other critical components. I advocate for extended practical training programmes, perhaps up to a year, to equip students with the necessary skills. Collaborations between academia and MSMEs can be particularly beneficial, facilitating internships and practical training that prepare students for real-world challenges. This partnership can significantly enhance the skilled workforce needed in the rapidly evolving EV industry.

Given that global players are far ahead in high-voltage battery technology, does India have sufficient time to catch up and develop in this area?

Yes, India can develop and advance in high-voltage battery technology despite the current global lead by other countries. India has significant market potential for high-voltage batteries, with projections suggesting a 40 to 50% market share in electric vehicles over the next 5 to 10 years. Several Indian companies are already making

strides in this area, like Tata with its buses and Nexon EVs, and Mahindra, which are contributing to the growth of this sector. While newer technologies such as sodium-ion and hydrogen fuel cells are emerging, they still require a few more years to mature. Meanwhile, lithium-ion technology remains a primary focus, and continued investment and innovation are crucial for catching up and potentially leading in specific segments.

What are five areas MSMEs should focus on within the high-voltage battery sector to capitalise on market opportunities?

MSMEs should concentrate on developing components where there is both a high demand and a technological gap that can be filled locally:

1. Wiring harnesses – Essential for connecting various electrical components in high-voltage systems
2. Cold plates – Crucial for thermal management in battery packs, ensuring batteries operate within safe temperature ranges
3. Bus bars – Conductors that distribute power across the battery cells, needing to handle high voltages efficiently
4. Lightweight enclosures – Such as those made from low-pressure die-cast aluminium, which provide robust protection for battery packs while maintaining lower weight to optimise energy density
5. Defence sector components – Given their strategic importance and mandated local production, developing battery technologies for defence applications can be a lucrative focus area.

By concentrating on these components, MSMEs can significantly contribute to the local ecosystem, reduce dependencies on imports, and align with national goals for self-reliance and technological advancement in the EV sector. **EFY**